

Exploratory and Predictive Analysis of Higher Education Institutions: Public vs. Private Classification

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Abstract

This report presents an exploratory, multivariate and predictive analysis of higher education institutions, with the objective of distinguishing public and private institutions using academic, financial and institutional indicators. The dataset used is the *US Colleges* dataset, based on data from *US News & World Report*, containing 977 higher education institutions and 26 predictor variables. The work follows a complete data analysis pipeline, beginning with data cleaning, treatment of missing values, transformation of skewed variables, feature engineering and standardisation. Exploratory Data Analysis is then conducted through histograms, density plots, boxplots, violin plots, scatterplots and a correlation matrix. Principal Component Analysis is used to study the multivariate structure of the dataset, while K-Means clustering is applied to identify natural groups among institutions. Finally, supervised classification models are compared, namely Linear Discriminant Analysis, Quadratic Discriminant Analysis, Logistic Regression and K-Nearest Neighbours. The results show that the distinction between public and private institutions is strongly related to tuition, institutional size, graduation rate, faculty qualification, number of applications and student enrolment. Among the supervised models, KNN achieved the best AUC-ROC value, followed by LDA, QDA and Logistic Regression.

1 Introduction

The main objective of this project is to analyse whether public and private higher education institutions can be distinguished using a set of academic, financial and institutional variables. The dataset contains information about colleges and universities, including variables related to applications, admissions, enrolment, tuition, graduation rate, faculty qualifications, expenditure and student-faculty ratio.

The research question guiding this work is:

Can academic, financial and institutional indicators be used to classify higher education institutions as public or private?

This question is relevant because the United States higher education system presents structural differences between public and private institutions. These differences may appear in terms of costs, institutional size, selectivity, academic performance, faculty profile and financial resources. Public institutions tend to have larger student populations and lower tuition values, while private institutions are often associated with higher tuition, smaller size and, in some cases, higher selectivity or graduation rates.

The analysis combines descriptive, exploratory, multivariate and predictive methods. First, the data are prepared and transformed. Then, Exploratory Data Analysis is used to understand the distribution of the variables and the differences between public and private institutions. Principal Component Analysis is applied to reduce dimensionality and interpret the main axes of variation.

K-Means clustering is used to investigate whether the natural structure of the data reflects the public/private distinction. Finally, supervised models are trained and compared in order to evaluate the predictive power of the variables.

The report follows the same methodological structure as the project presentation: dataset description, preprocessing, exploratory analysis, PCA, clustering, supervised models, model evaluation and conclusions.

2 Dataset Description

The dataset used in this project is the *US Colleges* dataset, originally based on data from *US News & World Report*. Each observation corresponds to one higher education institution. The dataset contains 977 institutions and 26 predictor variables describing academic, financial and institutional characteristics.

The response variable is the type of institution, classified as either:

- **Public:** public higher education institution;
- **Private:** private higher education institution.

The target distribution is imbalanced, with approximately 30% public institutions and 70% private institutions. This imbalance is relevant because model evaluation should not rely only on Accuracy. Metrics such as AUC-ROC, Balanced Accuracy, Sensitivity, Specificity and F1-score are more informative in this context.

The variables used in the analysis include original numerical variables and derived indicators. The most relevant variables are presented in Table 1.

Table 1: Main variables used in the analysis.

Variable	Description
log_applications_received	Logarithm of the number of applications received by the institution.
log_students_enrolled	Logarithm of the number of students enrolled.
log_tuition_outstate	Logarithm of out-of-state tuition.
pc_alumni_donors	Percentage of alumni who donate to the institution.
pc_faculty_with_phd	Percentage of faculty members with a PhD.
pc_faculty_with_terminal_degree	Percentage of faculty members with a terminal degree.
pc_graduation_rate	Graduation rate of the institution.
tx_acceptance_rate	Proportion of applications accepted.
tx_yield_rate	Proportion of accepted students who enrol.
vl_room_board	Cost of room and board.
vl_student_faculty_ratio	Student-faculty ratio.
vl_expenditure_per_student	Expenditure per student.
qt_undergraduate_students	Number of undergraduate students.
qt_postgraduate_students	Number of postgraduate students.

The variables represent several institutional dimensions: cost, institutional size, selectivity, academic quality, faculty profile, expenditure per student and attractiveness of the institution.

3 Data Preparation

Data preparation was an essential stage of the project. Before applying exploratory or predictive methods, the dataset was inspected and transformed to improve the quality and reliability of the analysis.

The preprocessing stage included the verification of impossible values, the identification of inconsistencies, the analysis of extreme outliers and the treatment of missing values. Missing values were imputed using the median, because the median is more robust than the mean in the presence of skewed distributions and extreme values. This decision is appropriate because several variables related to institutional size and costs present strong asymmetry.

3.1 Derived Variables

Two derived variables were created: acceptance rate and yield rate. The acceptance rate measures the proportion of applicants accepted by the institution:

$$\text{Acceptance Rate} = \frac{\text{Applications Accepted}}{\text{Applications Received}} \quad (1)$$

The yield rate measures the proportion of accepted students who effectively enrol:

$$\text{Yield Rate} = \frac{\text{Students Enrolled}}{\text{Applications Accepted}} \quad (2)$$

These variables are important because they allow institutions of different sizes to be compared using relative measures instead of absolute counts. For example, a large public university may receive many more applications than a small private college, but this does not necessarily mean that it is more selective. The acceptance rate provides a more interpretable measure of selectivity, while the yield rate can be interpreted as an indicator of institutional attractiveness among admitted students.

3.2 Logarithmic Transformations

Some variables showed strong asymmetry and the presence of extreme values. This is particularly common in variables related to applications, enrolment and tuition. To reduce the influence of extreme values and improve visual interpretation, logarithmic transformations were applied.

The transformation used was:

$$x_{\log} = \log(1 + x) \quad (3)$$

The addition of 1 avoids computational problems when the original variable contains zero values. The main transformed variables were:

- `log_applications_received;`
- `log_students_enrolled;`
- `log_tuition_outstate.`

These transformations were especially important for PCA, K-Means and KNN, because these methods can be strongly affected by variables with large scales or extreme observations.

3.3 Standardisation

Standardisation was applied before the multivariate and distance-based methods. This step is required because PCA, K-Means and KNN are sensitive to the scale of the variables. Without standardisation, variables with larger numerical ranges would dominate the analysis.

The standardisation formula is:

$$z = \frac{x - \bar{x}}{s} \quad (4)$$

where $\bar{x}$ is the mean and s is the standard deviation of the variable.

4 Exploratory Data Analysis

Exploratory Data Analysis was used to understand the behaviour of the variables and identify differences between public and private institutions. The analysis includes histograms, density plots, boxplots, violin plots, scatterplots and a correlation matrix.

4.1 Histograms

Figure 1 presents histograms of selected numerical variables by institution type.

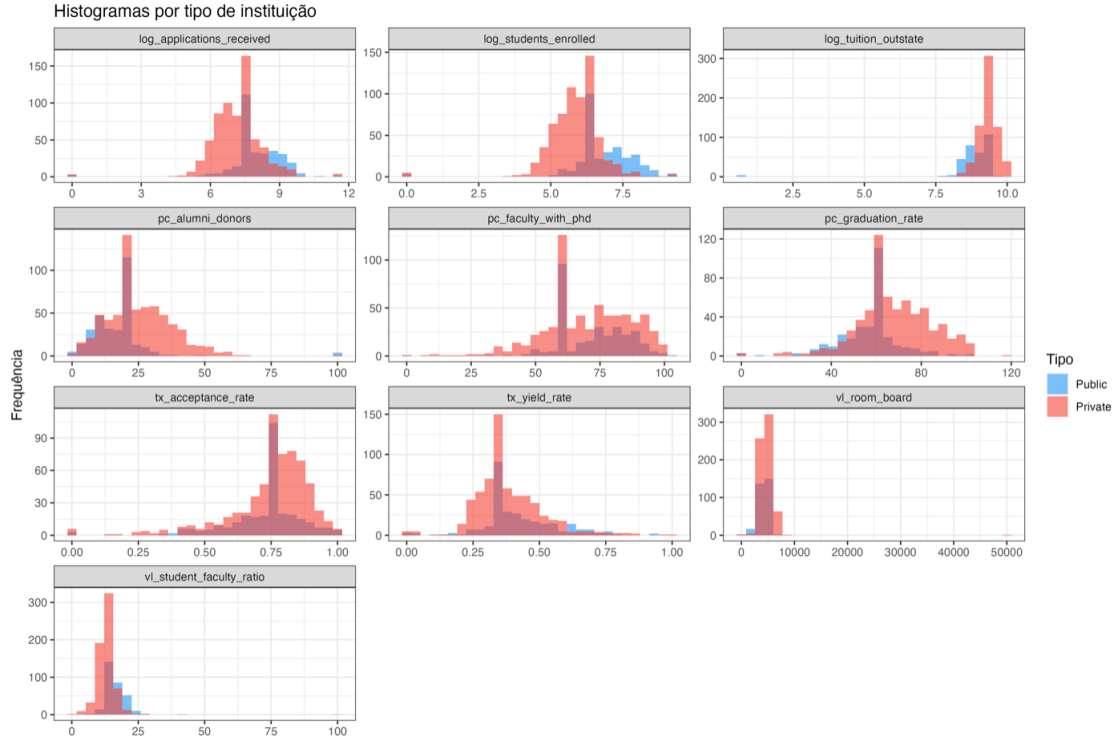


Figure 1: Histograms of selected variables by institution type.

The histograms show that public and private institutions differ in several variables. For `log_applications_received` and `log_students_enrolled`, public institutions tend to appear more frequently in higher ranges, suggesting that they generally receive more applications and enrol more students. This is consistent with the expected larger scale of public universities.

For `log_tuition_outstate`, private institutions are more concentrated at higher values. This indicates that tuition is one of the variables that most clearly distinguishes the two groups. Graduation rate also shows relevant differences, with private institutions generally presenting higher values and greater dispersion.

The histograms also reveal the presence of outliers and asymmetry in some variables, which supports the decision to apply logarithmic transformations to selected variables.

4.2 Density Plots

Figure 2 presents density plots for the same set of variables.

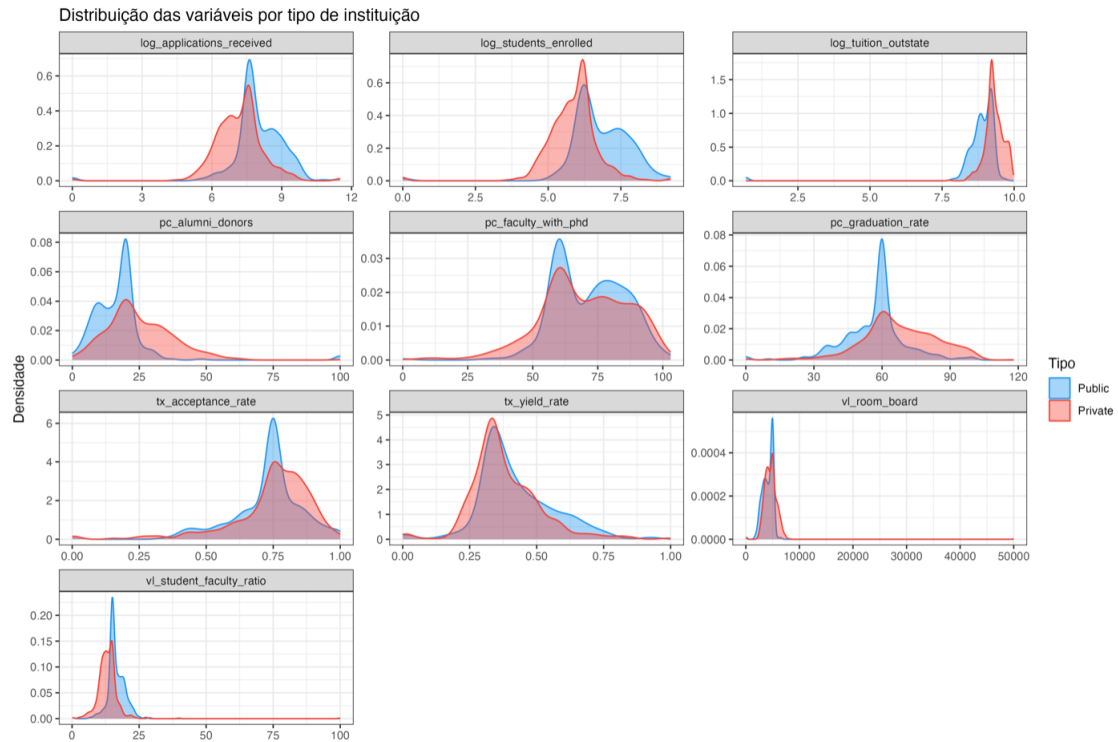


Figure 2: Density distributions of selected variables by institution type.

The density plots allow a clearer comparison between the two groups. When the curves of public and private institutions overlap, the variable has limited discriminative power. When the curves are separated, the variable contributes more strongly to distinguishing the groups.

The clearest differences appear in variables related to tuition, applications, enrolment and graduation rate. Public institutions show higher density in larger enrolment values, while private institutions show higher density in tuition and graduation rate. The acceptance rate and yield rate distributions overlap more substantially, suggesting that these variables alone may not be sufficient to classify institutions, although they still provide useful information when combined with other predictors.

4.3 Boxplots

Figure 3 presents boxplots by institution type.

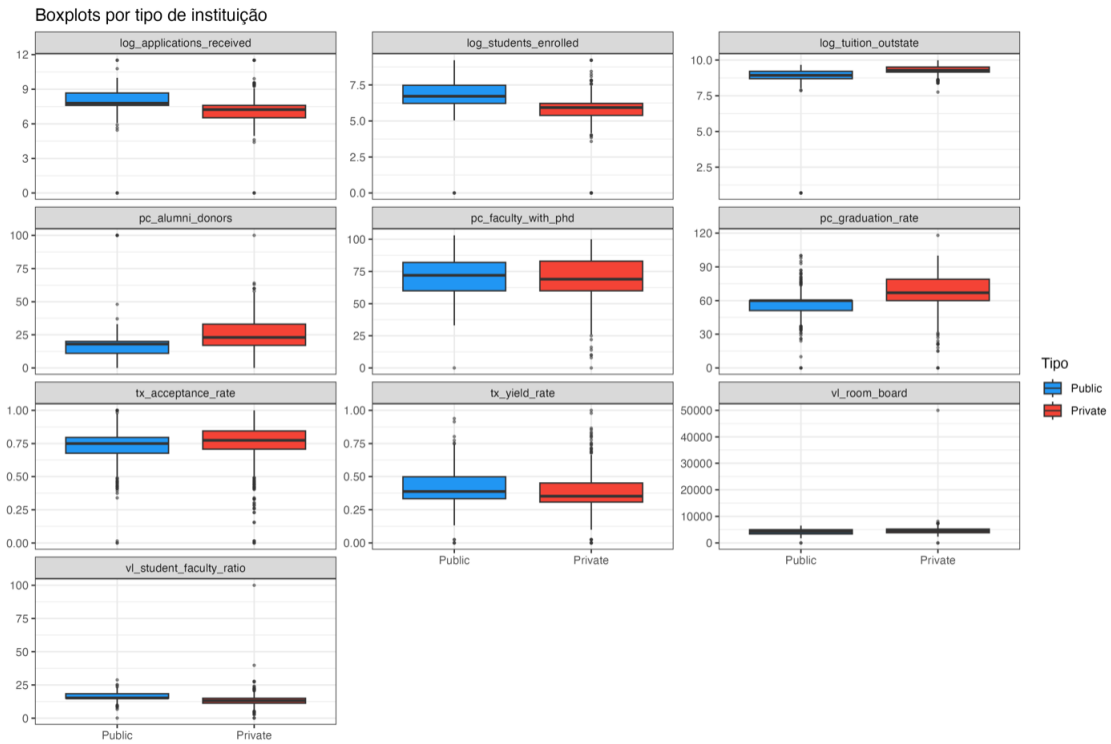


Figure 3: Boxplots of selected variables by institution type.

The boxplots reinforce the conclusions obtained from the histograms and density plots. Public institutions tend to have higher median values for applications received and students enrolled, confirming their larger institutional scale. Private institutions tend to have higher values for tuition and graduation rate.

The boxplots also highlight the existence of outliers. For example, some institutions show unusually high values in room and board, student-faculty ratio or alumni donors. These values were not removed automatically because they may represent genuine institutional profiles. Instead, the analysis used transformations and robust interpretation to reduce their influence.

4.4 Violin Plots

Figure 4 presents violin plots by institution type.

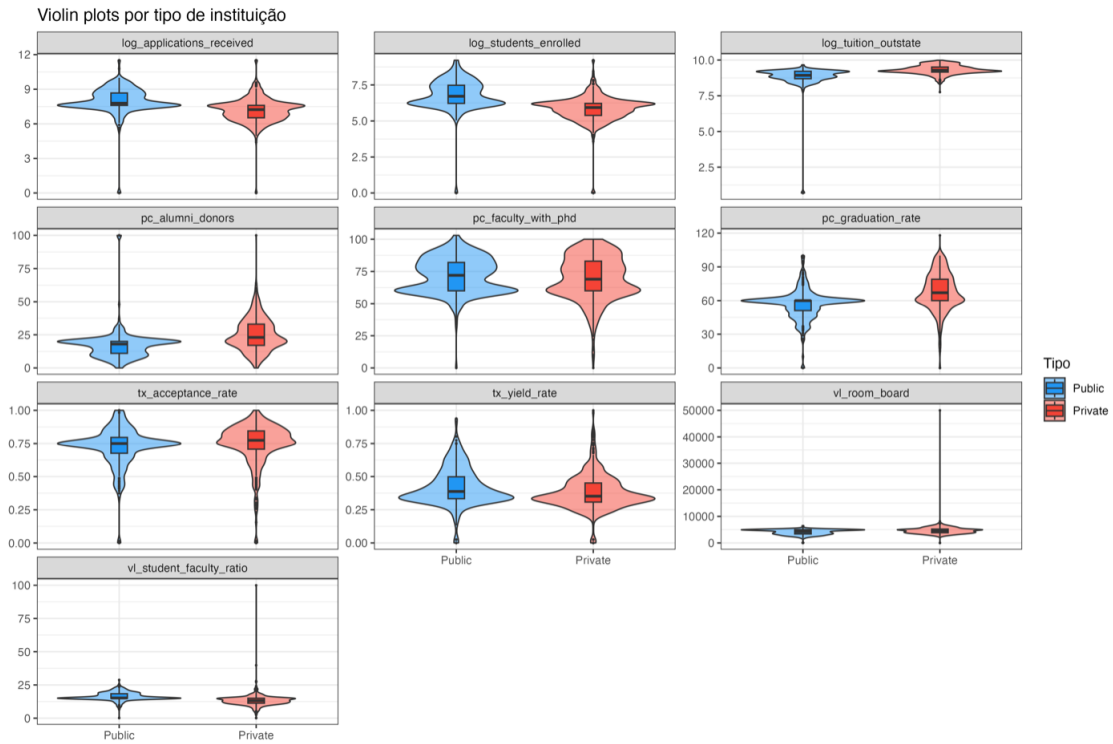


Figure 4: Violin plots of selected variables by institution type.

The violin plots combine information from boxplots and density plots. They show both the distribution shape and the central tendency of each variable. These plots confirm that public institutions are generally larger in terms of applications and enrolment, while private institutions tend to present higher tuition and higher graduation rates.

The shape of the violins also reveals that private institutions are more heterogeneous in some variables, especially graduation rate and alumni donors. This means that private institutions do not form a perfectly homogeneous group. Some private institutions are highly selective and expensive, while others have profiles closer to public institutions.

4.5 Acceptance Rate and Yield Rate

Figure 5 shows the relationship between acceptance rate and yield rate.

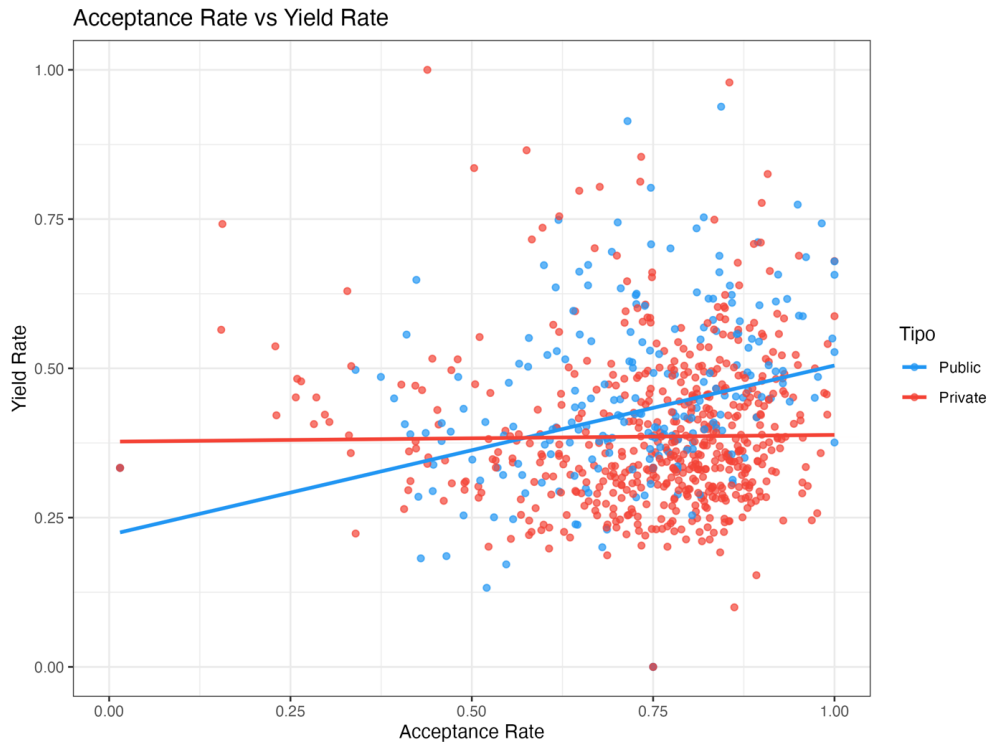


Figure 5: Relationship between acceptance rate and yield rate.

The scatterplot shows a considerable overlap between public and private institutions. This indicates that acceptance rate and yield rate alone do not fully separate the two groups. However, the trend lines suggest different patterns.

For public institutions, the relationship between acceptance rate and yield rate appears more positive. This may indicate that, among public institutions, higher acceptance rates are associated with higher enrolment among accepted students. For private institutions, the trend is flatter, suggesting that yield rate varies less directly with acceptance rate.

This result suggests that selectivity and enrolment behaviour differ between the two types of institutions, but the relationship is not strong enough to classify institutions on its own.

4.6 Tuition and Graduation Rate

Figure 6 presents the relationship between log tuition and graduation rate.

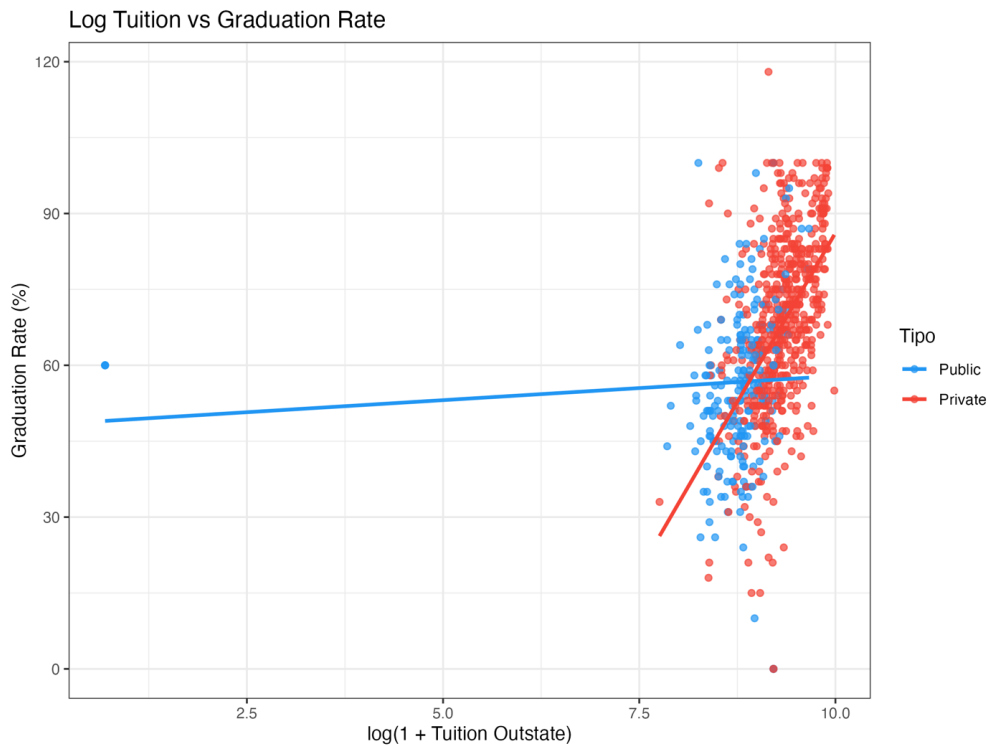


Figure 6: Relationship between log tuition and graduation rate.

This figure shows a clearer distinction between public and private institutions. Private institutions are concentrated at higher tuition values and tend to show a positive relationship between tuition and graduation rate. This suggests that, among private institutions, higher tuition may be associated with better academic outcomes or with institutional characteristics linked to higher graduation rates.

For public institutions, the trend is much weaker. Public institutions are concentrated in lower tuition ranges, and the relationship between tuition and graduation rate appears flatter. This reinforces the idea that tuition is one of the most important variables for distinguishing public and private institutions.

However, this relationship should not be interpreted as causal. Higher tuition does not necessarily cause higher graduation rates. It may be associated with other factors such as selectivity, institutional resources, academic support or student socioeconomic background.

4.7 Correlation Matrix

Figure 7 presents the correlation matrix of the numerical variables.

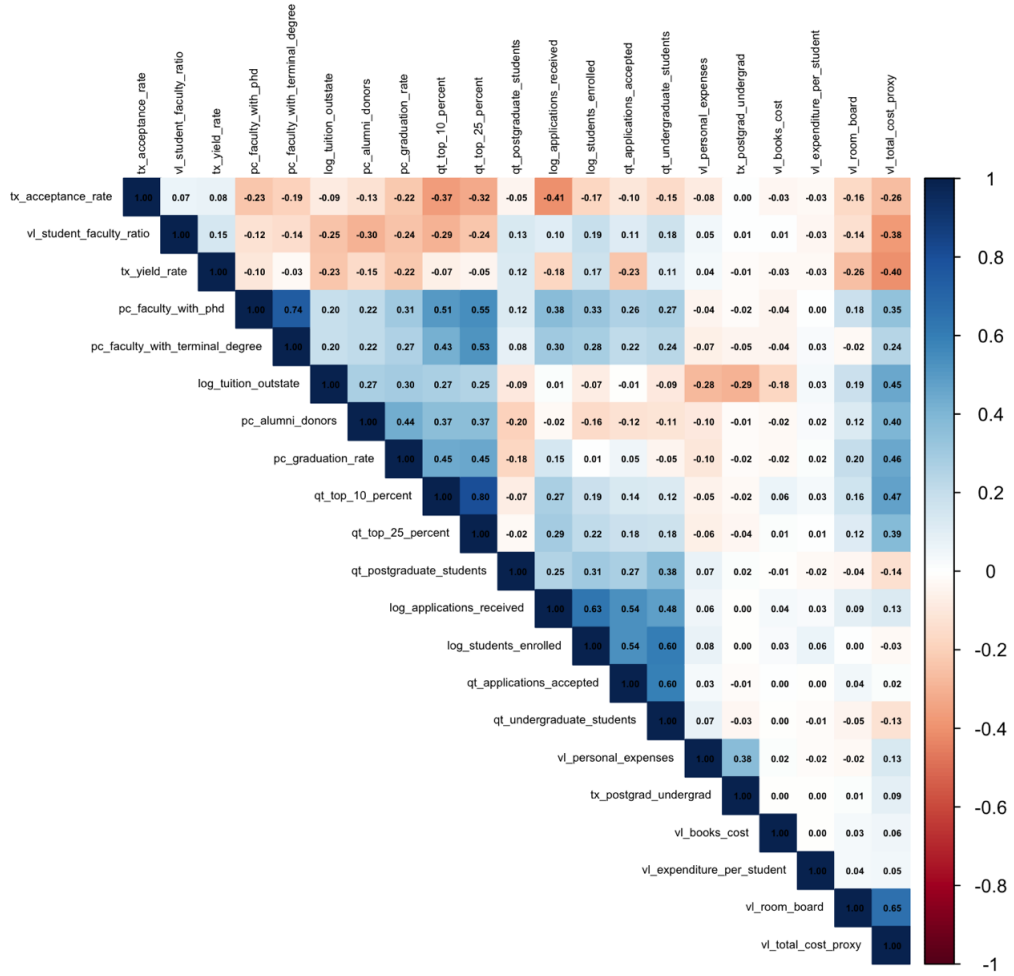


Figure 7: Correlation matrix of numerical variables.

The correlation matrix reveals several important relationships. Variables associated with institutional size, such as applications received, applications accepted and students enrolled, are positively correlated. This is expected because institutions that receive more applications also tend to accept and enrol more students.

There are also positive correlations between variables related to academic profile, such as the percentage of students in the top 10% and top 25%, graduation rate and faculty qualifications. Tuition is positively associated with total cost and, to some extent, with graduation rate and alumni donors.

The existence of correlated variables justifies the use of Principal Component Analysis. PCA is useful in this context because it summarises correlated information into a smaller number of uncorrelated components. The matrix also indicates possible multicollinearity, which may affect the stability of coefficients in Logistic Regression and discriminant analysis models.

5 Principal Component Analysis

Principal Component Analysis was applied to understand the multivariate structure of the dataset and reduce dimensionality. Since the variables are measured in different units, PCA was performed after standardisation.

5.1 Scree Plot

Figure 8 presents the scree plot.

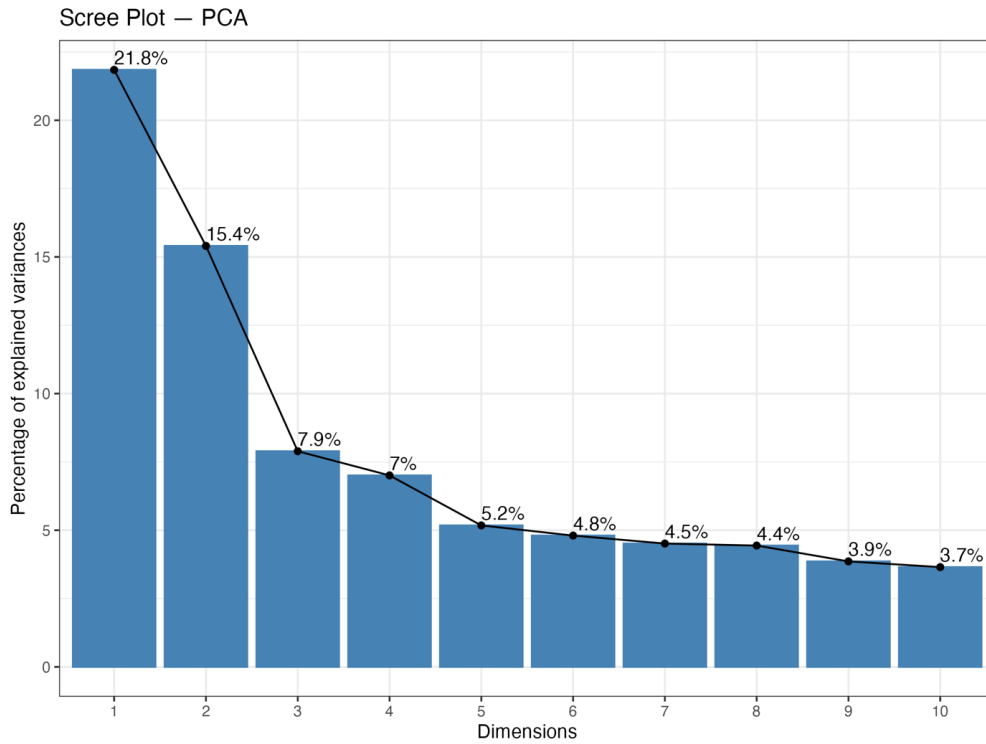


Figure 8: Scree plot of the Principal Component Analysis.

The scree plot shows that the first principal component explains 21.8% of the total variance, while the second component explains 15.4%. Together, the first two components explain 37.2% of the variance. The third and fourth components explain 7.9% and 7.0%, respectively.

The decrease after the first two components suggests that the first dimensions capture the most important structure in the dataset. However, the explained variance is distributed across several components, which indicates that the dataset is multidimensional and cannot be fully summarised by only one or two variables. Approximately six principal components are needed to explain a large proportion of the total variance, reinforcing the multidimensional nature of the phenomenon.

5.2 PCA Individual Map

Figure 9 presents the projection of institutions on the first two principal components.

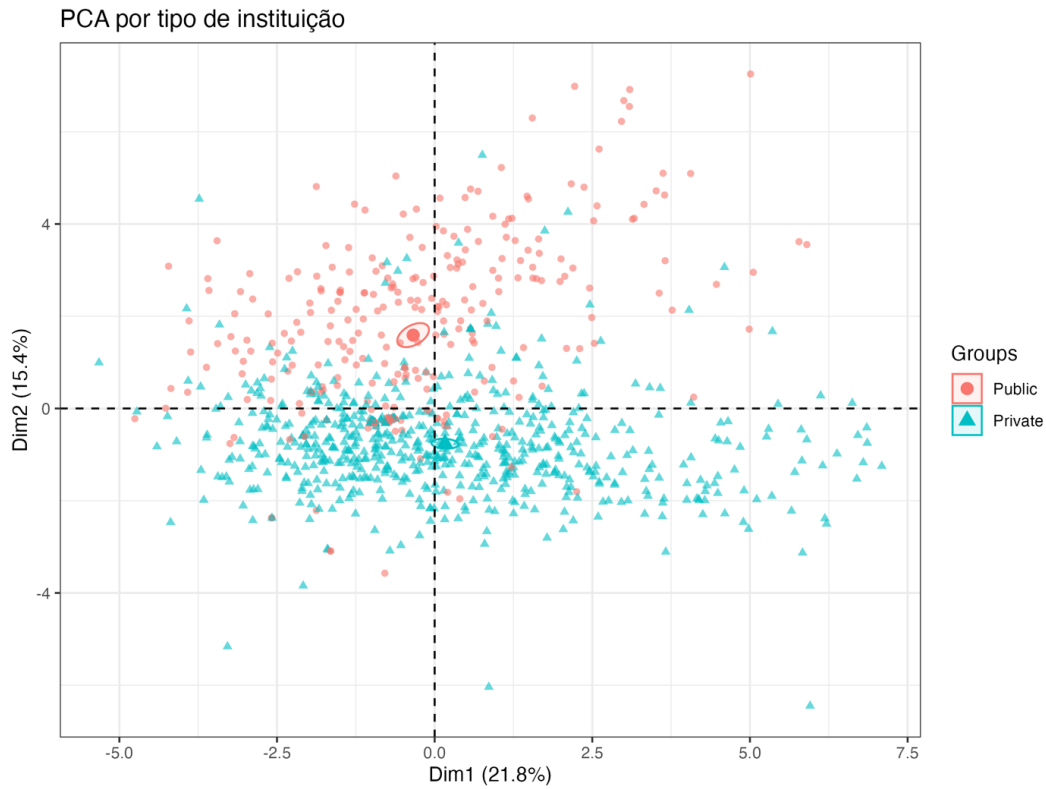


Figure 9: PCA individual map by institution type.

The PCA individual map shows partial separation between public and private institutions. Public institutions appear more homogeneous, while private institutions show greater dispersion and heterogeneity. However, there is still considerable overlap between the two groups.

This result indicates that the public/private distinction is reflected in the multivariate structure of the data, but it is not absolute. Some private institutions have characteristics similar to public institutions, and some public institutions share characteristics with private institutions. Therefore, classification requires a combination of variables and cannot be based only on a simple visual separation.

5.3 PCA Variable Contributions

Figure 10 presents the contribution of variables to the first two principal components.

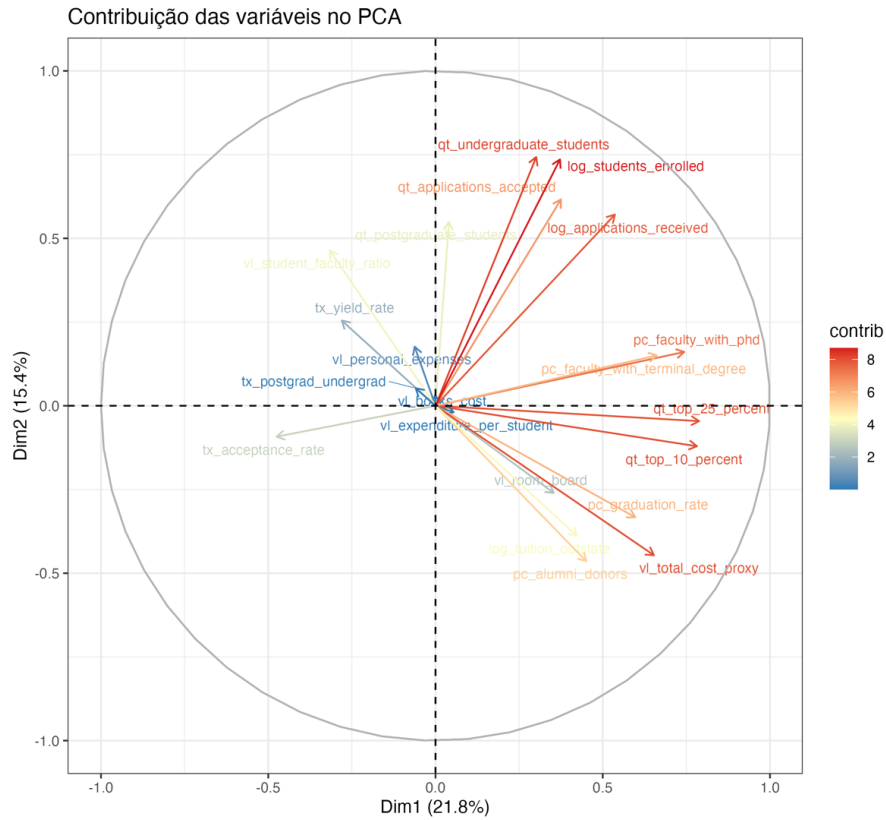


Figure 10: Contribution of variables to the PCA.

The variable map helps interpret the meaning of the principal components. Variables pointing in the same direction are positively correlated, while variables pointing in opposite directions are negatively correlated.

The first principal component is mainly associated with institutional scale. Variables such as applications received, applications accepted, undergraduate students and students enrolled contribute to this dimension. Therefore, PC1 can be interpreted as a size or volume axis, separating institutions according to their institutional scale.

The second principal component is more closely related to prestige, selectivity and academic quality. Variables such as graduation rate, top 10% and top 25% students, faculty qualifications and tuition contribute to this dimension. Acceptance rate appears in the opposite direction to variables associated with academic prestige, suggesting that more selective institutions tend to present lower acceptance rates and stronger academic indicators.

This interpretation confirms that the distinction between public and private institutions depends on several dimensions: scale, selectivity, academic quality and financial profile.

6 Clustering Analysis

K-Means clustering was applied to identify natural groups of institutions without using the public/private label during model fitting. The goal was to understand whether the internal structure of the data corresponds to the administrative classification of institutions.

6.1 Elbow Method

Figure 11 presents the elbow method.

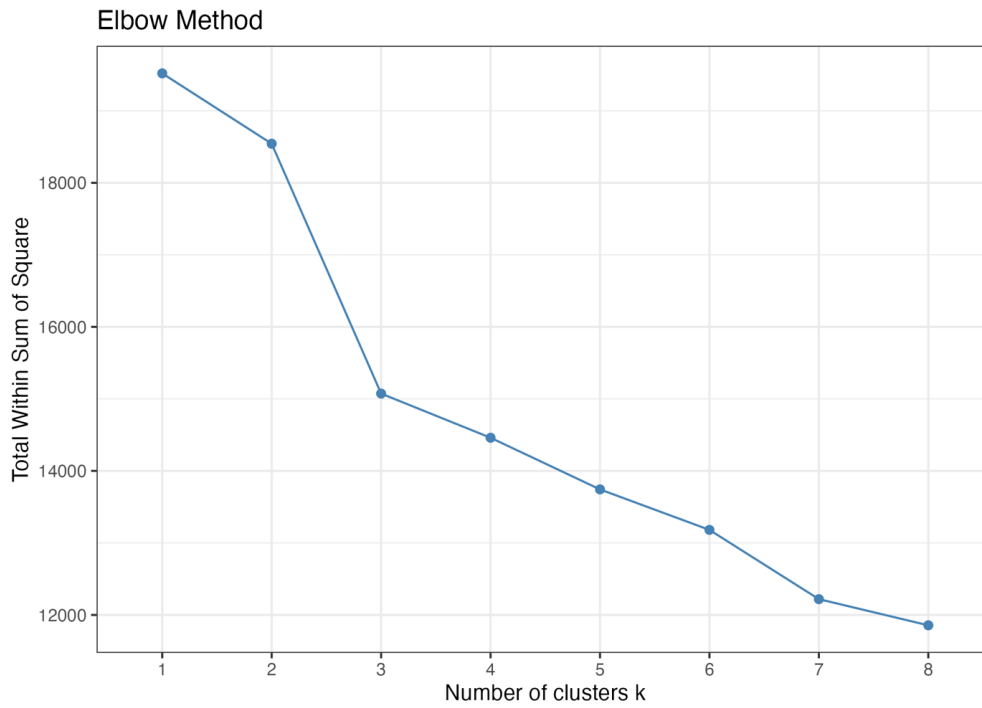


Figure 11: Elbow method for K-Means clustering.

The elbow method shows a strong reduction in within-cluster sum of squares when moving from one to two or three clusters. After this point, the improvement becomes more gradual. This suggests that a small number of clusters may be sufficient to capture the main structure of the data.

Although the public/private target has two classes, the elbow plot suggests that the data may contain more than two natural groups. This is important because public and private institutions may include internal subgroups with different profiles.

6.2 Silhouette Method

Figure 12 presents the silhouette method.

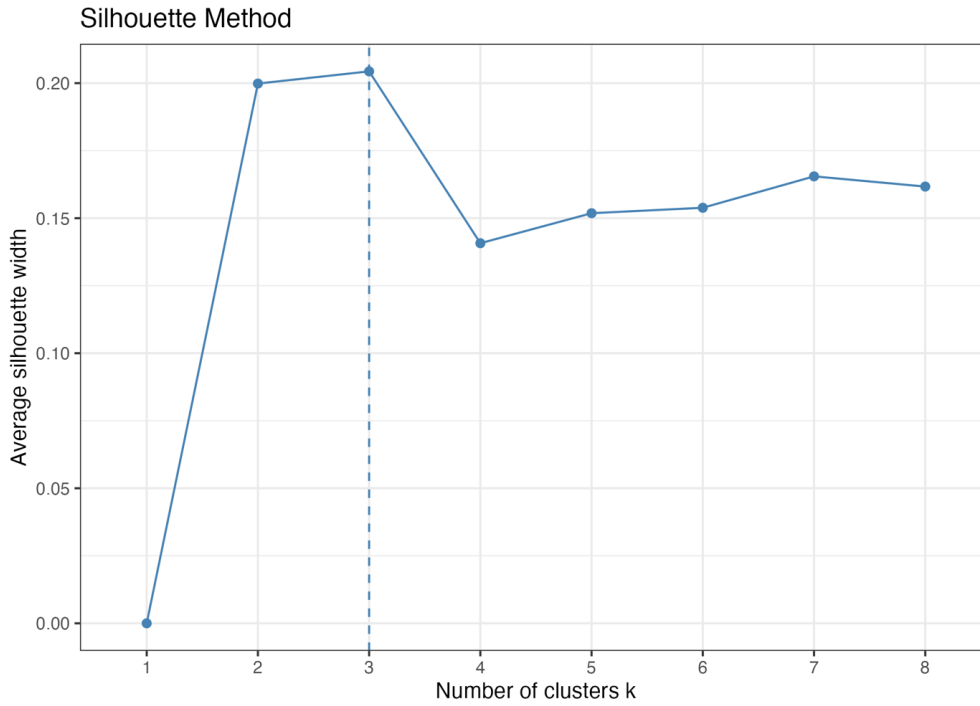


Figure 12: Silhouette method for K-Means clustering.

The silhouette method reaches its highest value at $k = 3$, suggesting that three clusters provide the best separation according to this criterion. However, the value for $k = 2$ is also relevant because it allows a direct comparison with the binary target variable, Public versus Private.

Therefore, $k = 2$ was used for interpretability and comparison with the target variable, while $k = 3$ was acknowledged as the best solution according to the silhouette criterion. This interpretation reconciles the statistical criterion with the substantive objective of the project.

6.3 K-Means with Two Clusters

Figure 13 presents the K-Means solution with $k = 2$.

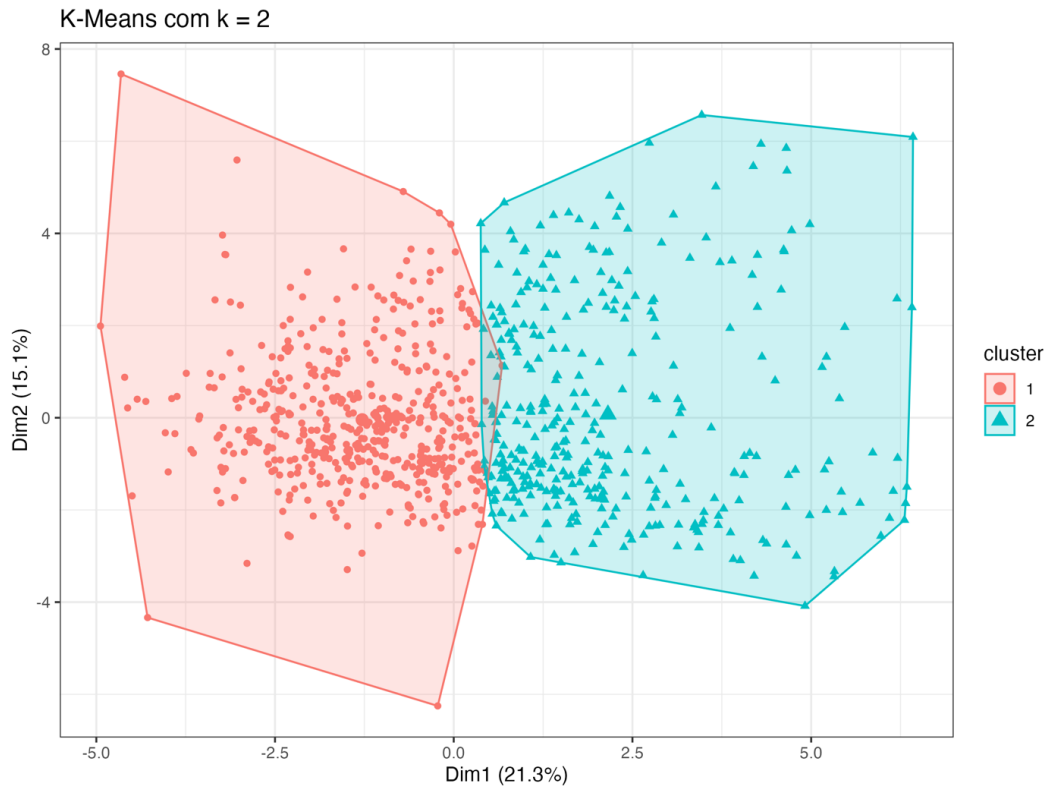


Figure 13: K-Means clustering with $k = 2$.

The two-cluster solution separates institutions mainly along the first PCA dimension. One cluster is concentrated on the left side of the plot, while the other is located more on the right. This suggests that the clustering is capturing broad differences in institutional profile.

The K-Means solution with $k = 2$ shows a partial separation between institutions, but not a perfect division between public and private institutions. One cluster tends to concentrate more private institutions, while the other presents a more mixed profile. This reinforces the idea that institutional type is not explained by a single dimension, but by a combination of institutional factors.

6.4 K-Means with Five Clusters

Figure 14 presents the K-Means solution with $k = 5$.

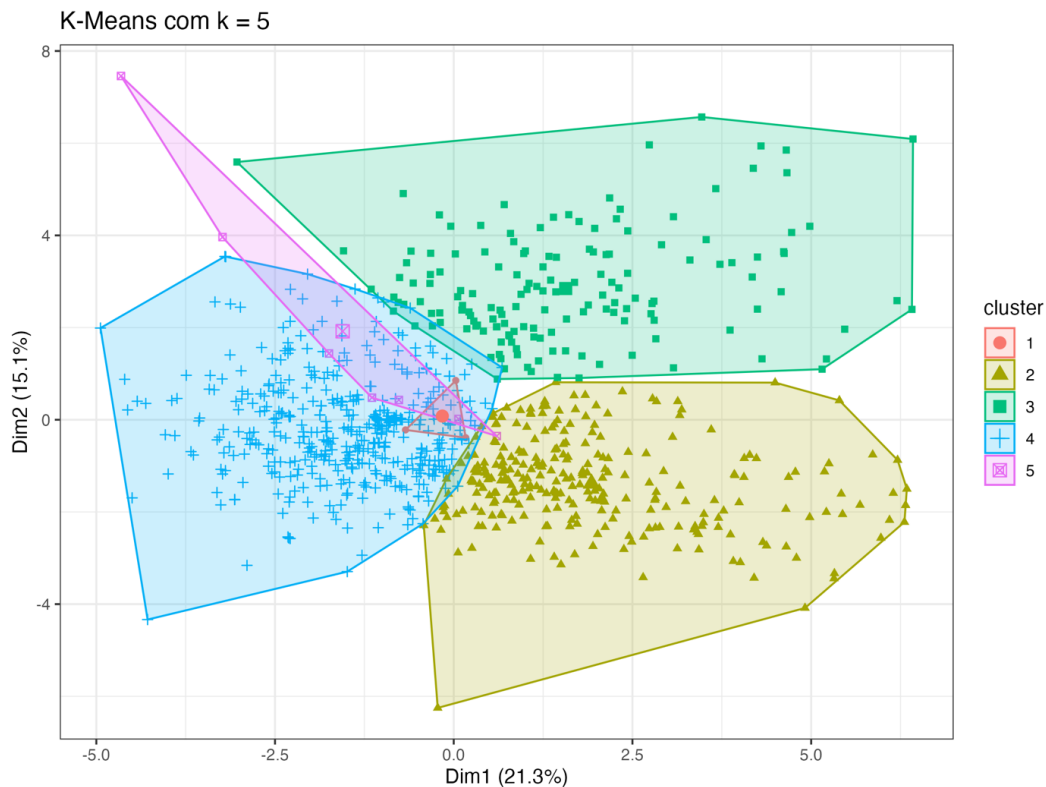


Figure 14: K-Means clustering with $k = 5$.

The five-cluster solution provides a more detailed view of the dataset. Instead of dividing institutions into only two broad groups, it identifies different institutional profiles. Some clusters are associated with larger institutions, while others seem to represent smaller or more selective institutions.

This result is important because it shows that the public/private distinction does not capture all the heterogeneity in the data. Within each type of institution, there are different profiles related to size, cost, selectivity and academic outcomes.

7 Supervised Classification Models

After the exploratory and unsupervised analysis, supervised classification models were applied to predict whether an institution is public or private.

The models used were:

- Linear Discriminant Analysis;
- Quadratic Discriminant Analysis;
- Logistic Regression;
- K-Nearest Neighbours.

These models were selected because they represent different approaches. LDA and Logistic Regression are more interpretable and linear. QDA is more flexible because it allows different covariance matrices for each class. KNN is non-parametric and can capture non-linear patterns.

7.1 Model Assumptions

Before interpreting the supervised models, some statistical assumptions were considered. Box's M test rejected the equality of covariance matrices between groups, which is relevant for LDA because

this method assumes equal covariance structures. Mardia's test also rejected multivariate normality, indicating that the assumption of multivariate normal distributions is not fully satisfied.

These results do not invalidate the analysis, but they require caution when interpreting LDA and QDA. They also justify the inclusion of more flexible or non-parametric approaches, such as KNN. In addition, the class imbalance between public and private institutions reinforces the need to evaluate the models using AUC-ROC and balanced metrics instead of relying only on Accuracy.

7.2 Linear Discriminant Analysis

Linear Discriminant Analysis aims to find a linear combination of variables that best separates the classes. Figure 15 presents the standardised LD1 coefficients.

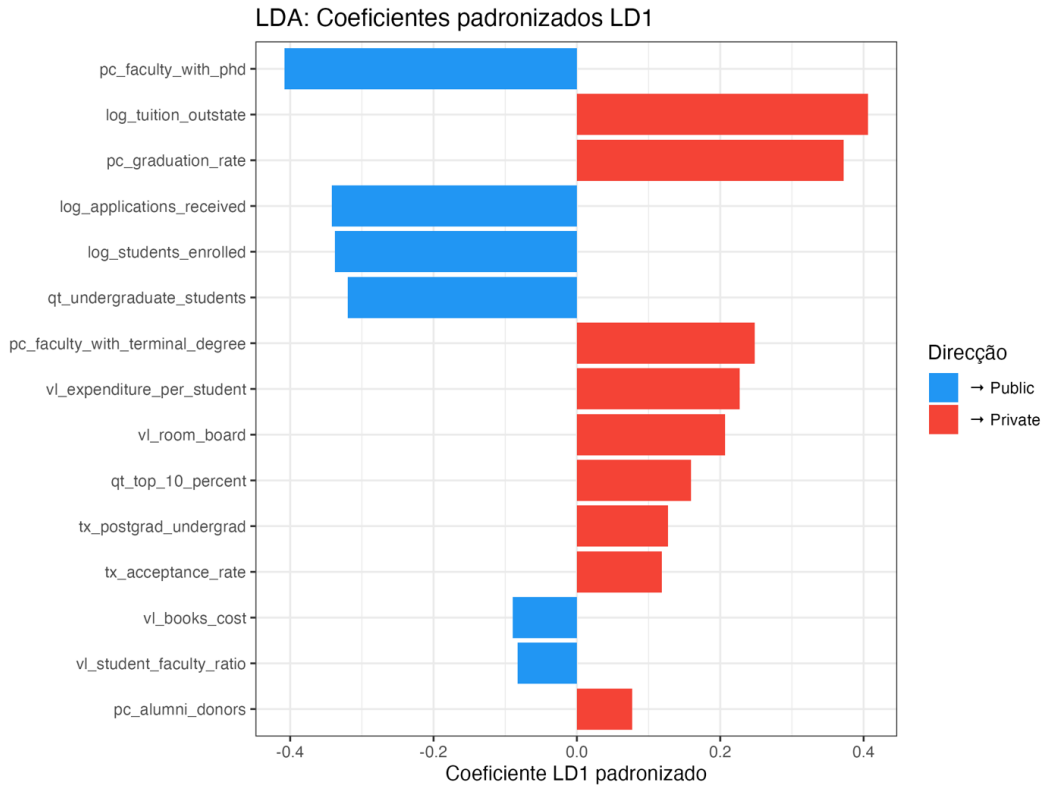


Figure 15: Standardised LD1 coefficients for Linear Discriminant Analysis.

The LDA coefficients indicate which variables contribute most strongly to the separation between public and private institutions. Positive coefficients point towards the private class, while negative coefficients point towards the public class.

The strongest variables associated with the private direction are `log_tuition_outstate`, `pc_graduation_rate`, `pc_faculty_with_terminal_degree`, `vl_expenditure_per_student` and `vl_room_board`. This suggests that private institutions are mainly characterised by higher tuition, higher graduation rates and higher academic or institutional resources.

The variables associated with the public direction include `pc_faculty_with_phd`, `log_applications_received`, `log_students_enrolled` and `qt_undergraduate_students`. This indicates that public institutions are more strongly associated with institutional size and volume.

7.3 LDA Variable Importance

Figure 16 presents the importance of variables in the LDA model using the absolute value of LD1 coefficients.

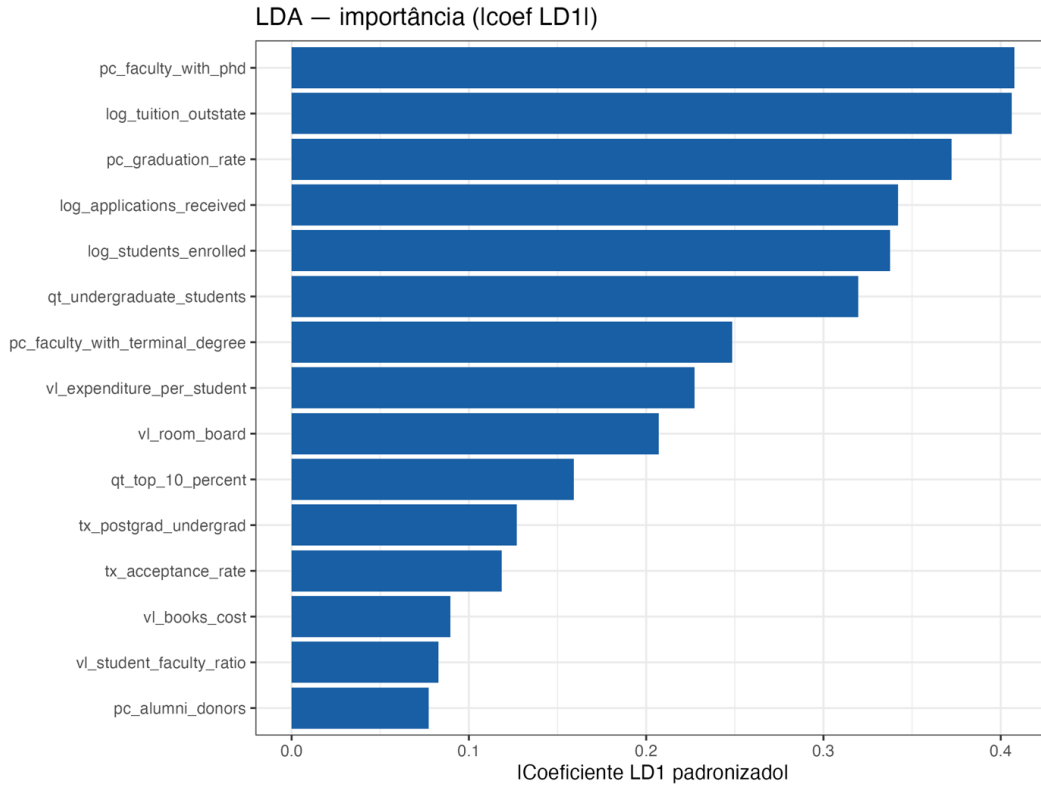


Figure 16: Variable importance in LDA based on absolute LD1 coefficients.

The most important variables in LDA are `pc_faculty_with_phd`, `log_tuition_outstate`, `pc_graduation_rate`, `log_applications_received`, `log_students_enrolled` and `qt_undergraduate_students`. This confirms that the classification is mainly driven by a combination of academic quality, cost and institutional size.

7.4 Logistic Regression

Logistic Regression was used as an interpretable probabilistic classification model. The model estimates the probability of an institution belonging to one of the two classes.

The general form of the model is:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)}} \quad (5)$$

where Y is the target variable and $X_1, \dots, X_p$ are the explanatory variables.

7.5 Logistic Regression Odds Ratios

Figure 17 presents the odds ratios of the significant variables in the Logistic Regression model.

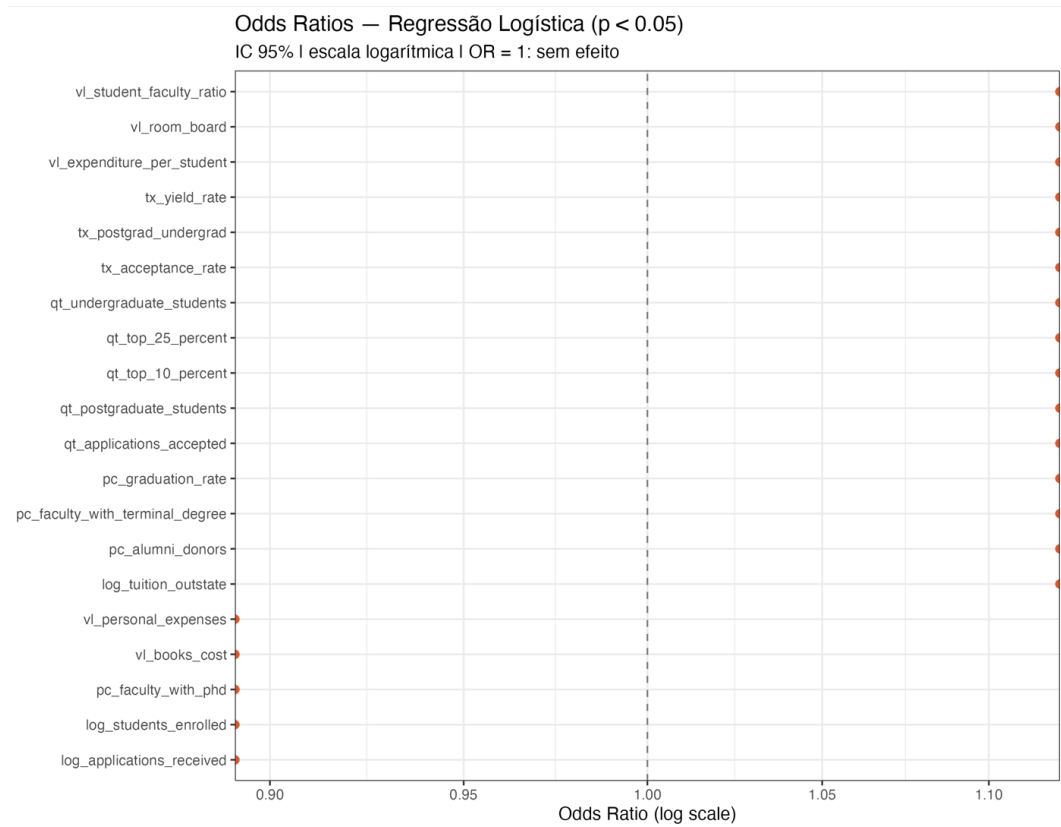


Figure 17: Odds ratios for Logistic Regression.

The odds ratio plot shows the effect of each variable on the odds of belonging to a class. An odds ratio greater than 1 indicates a positive association with the class coded as Private, while an odds ratio below 1 favours the Public class. The vertical reference line at $OR = 1$ represents no effect.

In this plot, several odds ratios are very close to the axis limits, suggesting that the model estimates strong effects for some variables. This may indicate that the variables are highly discriminative, but it may also reflect instability caused by multicollinearity or strong scaling effects. Therefore, the results should be interpreted together with the variable importance plot and the model performance metrics.

7.6 Logistic Regression Variable Importance

Figure 18 shows the variable importance in Logistic Regression based on the absolute value of the z-statistic.

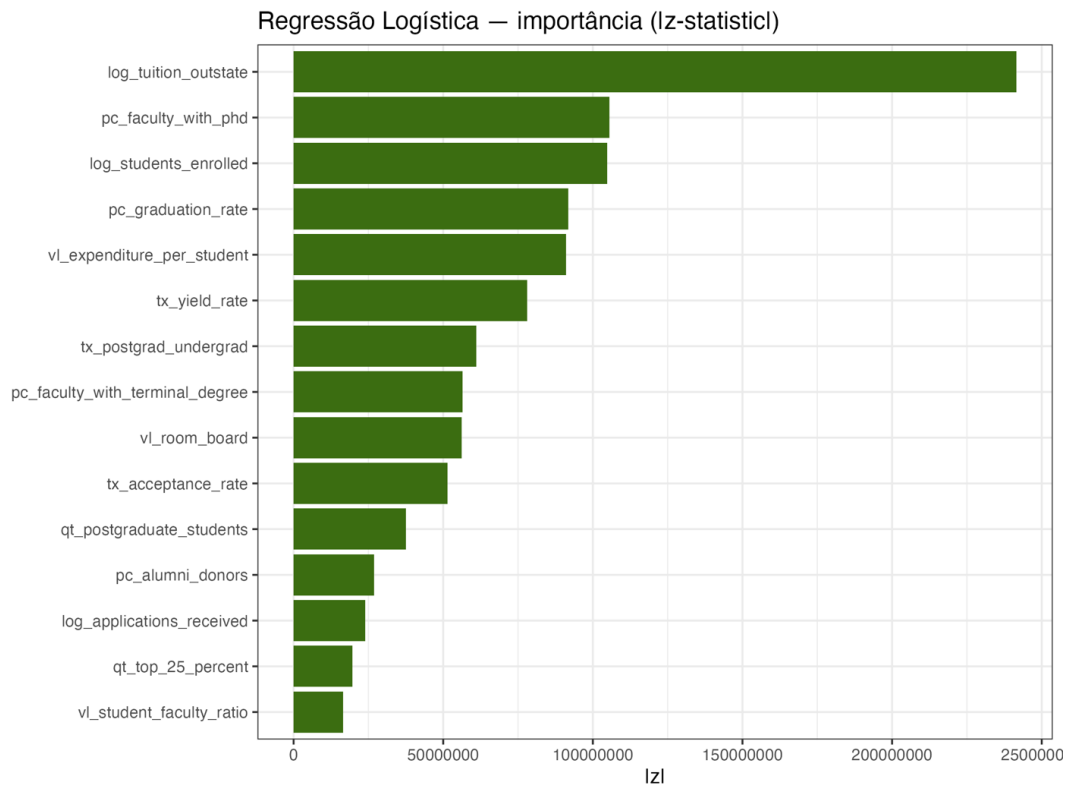


Figure 18: Variable importance in Logistic Regression based on absolute z-statistics.

The most important variable in Logistic Regression is `log_tuition_outstate`. Other relevant variables include `pc_faculty_with_phd`, `log_students_enrolled`, `pc_graduation_rate` and `vl_expenditure_per_student`. These results are consistent with the previous analyses: tuition, size, academic quality and expenditure are central variables in distinguishing public and private institutions.

7.7 K-Nearest Neighbours

K-Nearest Neighbours is a non-parametric classification method based on distance. The predicted class of an observation is determined by the majority class among its k nearest neighbours.

Since KNN is distance-based, standardisation was essential. Without standardisation, variables with larger numerical ranges would dominate the distance calculation. KNN is particularly relevant in this project because it can capture local and non-linear relationships.

7.8 Selection of K

Figure 19 presents the selection of the value of k using 10-fold cross-validation.

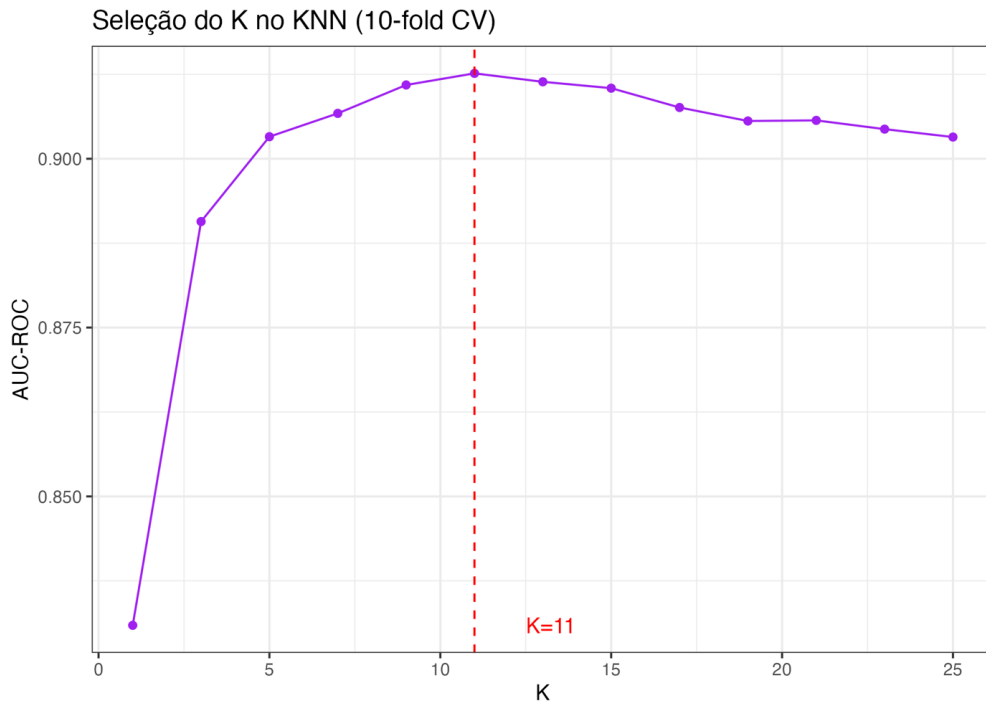


Figure 19: Selection of k in KNN using 10-fold cross-validation.

The graph shows that the best value is $k = 11$, which achieved the highest AUC-ROC during cross-validation. Smaller values of k may lead to overfitting, while larger values may oversmooth the decision boundary. The value $k = 11$ provides a balance between flexibility and generalisation.

7.9 KNN Variable Importance

Figure 20 presents the variable importance for the KNN model using univariate AUC.

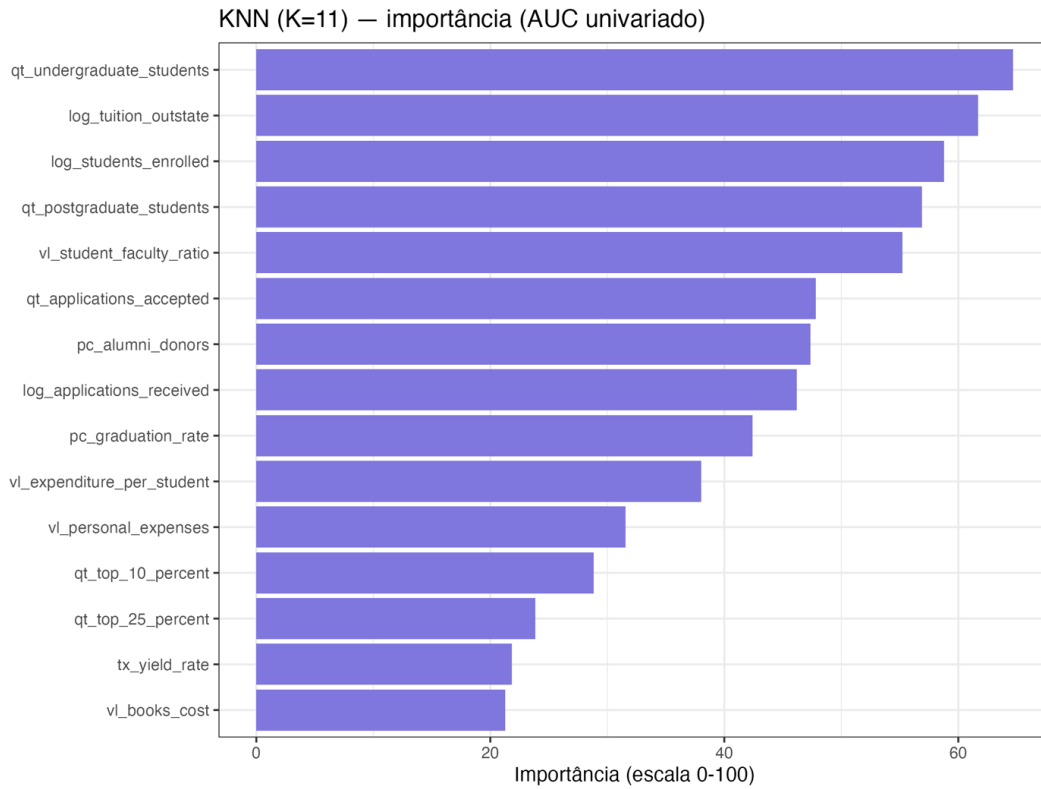


Figure 20: Variable importance in KNN using univariate AUC.

The KNN variable importance plot shows that `qt_undergraduate_students`, `log_tuition_outstate`, `log_students_enrolled`, `qt_postgraduate_students` and `vl_student_faculty_ratio` are among the most relevant variables.

This confirms that KNN is mainly using institutional size and tuition-related variables to distinguish between public and private institutions. The importance of undergraduate and postgraduate student counts suggests that scale is a central dimension in the classification.

8 Model Evaluation

The supervised models were evaluated using ROC curves and AUC values. ROC curves show the trade-off between sensitivity and specificity for different classification thresholds.

8.1 ROC Curves

Figure 21 presents the ROC curves for the four models.

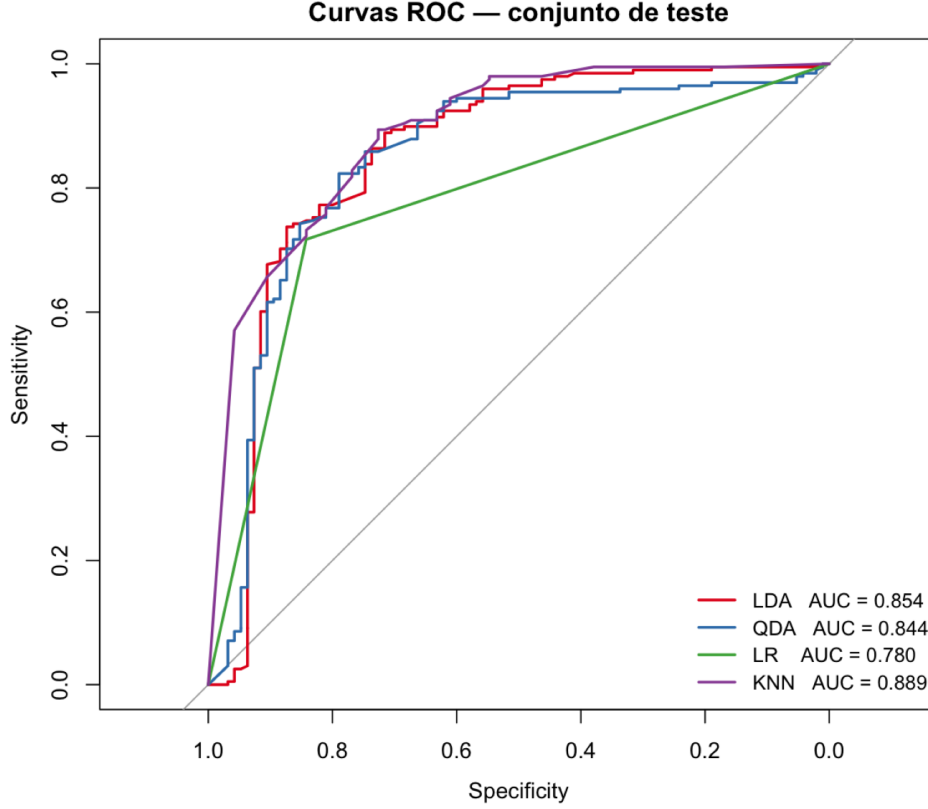


Figure 21: ROC curves for the supervised classification models on the test set.

All models perform better than a random classifier, represented by the diagonal line. The closer the curve is to the upper-left corner, the better the model is at distinguishing between public and private institutions.

The AUC values obtained were:

Table 2: AUC-ROC values for the supervised models.

Model	Type	AUC-ROC
KNN	Non-linear	0.889
LDA	Linear	0.854
QDA	Quadratic	0.844
Logistic Regression	Linear	0.780

The best model according to AUC-ROC is KNN, with an AUC of 0.889. This suggests that non-linear patterns are relevant in the classification of institutions. LDA also performs well, with an AUC of 0.854, showing that a linear combination of variables is already capable of separating the groups reasonably well. QDA obtains a similar result to LDA, while Logistic Regression has the lowest AUC among the four models.

Although KNN achieved the best predictive performance, LDA remains an important model because it offers a better balance between performance and interpretability. Logistic Regression is useful for probabilistic interpretation and odds ratios, but it was less competitive in terms of AUC.

9 Discussion

The results show that the distinction between public and private institutions is strongly associated with a combination of institutional size, tuition, academic outcomes and faculty characteristics.

Public institutions are generally associated with larger scale. They tend to receive more applications, enrol more students and have larger undergraduate populations. This pattern is visible in the histograms, density plots, boxplots, violin plots, PCA and LDA coefficients.

Private institutions are more strongly associated with higher tuition, higher graduation rates, room and board costs, expenditure per student and some indicators of academic profile. This is consistent with the expected differences between public and private higher education institutions.

The PCA confirmed that the dataset has a multidimensional structure. The first component is mainly related to institutional scale, while the second component is associated with prestige, selectivity and academic quality. The PCA individual map shows partial separation between public and private institutions, but also considerable overlap. This means that the distinction is real but not absolute.

The clustering analysis also supports this interpretation. The two-cluster solution captures a broad separation, but the silhouette method suggests that three clusters may better describe the natural structure of the data. The five-cluster solution reveals additional heterogeneity, suggesting that public and private institutions contain different subgroups.

The supervised models confirm that the variables have predictive power. KNN achieved the best AUC-ROC, suggesting that the relationship between predictors and institutional type is not purely linear. However, LDA remains a strong and interpretable model. Logistic Regression is useful for interpretation, but its lower AUC suggests that it may not capture all the relevant structure in the data.

10 Limitations

This analysis has some limitations. First, the dataset is relatively old and may not fully represent the current structure of higher education institutions. Public and private institutions may have changed substantially over time in terms of tuition, admissions, funding and student profile.

Second, the target variable is imbalanced, with approximately 70% private institutions and 30% public institutions. Although AUC-ROC and balanced metrics were considered, the class distribution may still influence the models.

Third, some statistical assumptions were violated. Box's M test rejected equality of covariance matrices and Mardia's test rejected multivariate normality. This affects the interpretation of LDA and QDA, even though these models can still be useful as comparative classifiers.

Fourth, several variables are correlated. Although this is addressed through PCA and careful interpretation, multicollinearity can affect models such as Logistic Regression and discriminant analysis.

Fifth, the classification problem is based on observational data. Therefore, the relationships identified should not be interpreted as causal. For example, higher tuition is associated with private institutions and higher graduation rates, but this does not mean that tuition directly causes better graduation outcomes.

Finally, KNN achieved the best predictive performance, but it is less interpretable than LDA or Logistic Regression. Therefore, there is a trade-off between predictive accuracy and interpretability.

11 Conclusion

This project investigated whether academic, financial and institutional indicators can be used to classify higher education institutions as public or private.

The exploratory analysis showed clear differences between the two groups. Public institutions tend to be larger, with more applications and higher enrolment. Private institutions tend to have higher tuition, higher graduation rates and higher values in several institutional quality indicators.

The PCA showed that the main structure of the data is explained by dimensions related to institutional scale, prestige, selectivity and academic quality. Although there is partial separation between public and private institutions, the overlap indicates that the distinction is not perfect.

The clustering analysis revealed that the natural structure of the data is more complex than a simple public/private division. The silhouette method suggested that three clusters may provide the best structure according to the average silhouette width, while the two-cluster solution was useful for comparison with the binary target. The five-cluster solution showed additional institutional profiles.

The supervised classification models confirmed that the dataset contains relevant predictive information. KNN achieved the best AUC-ROC value, followed by LDA, QDA and Logistic Regression. The most important variables across the models were tuition, student enrolment, number of undergraduate students, graduation rate, faculty qualifications and institutional expenditure.

Overall, the results support the conclusion that public and private institutions can be distinguished using academic, financial and institutional indicators, although the separation is not absolute and requires a multivariate approach.

12 Future Work

Future work could extend this analysis in several directions. First, more advanced machine learning models could be tested, such as Random Forest and XGBoost, in order to evaluate whether ensemble methods improve predictive performance. Second, regularisation methods such as LASSO and Ridge could be applied to reduce the impact of multicollinearity and improve model stability.

Since the target variable is imbalanced, future work could also explore SMOTE or class weighting strategies. These approaches may improve the ability of the models to correctly classify the minority class. Another relevant extension would be to validate the analysis using a more recent dataset, since the current data may not fully reflect the present structure of higher education institutions.

Finally, model interpretability could be improved using SHAP values or similar explainability techniques. This would allow a more detailed understanding of how each variable contributes to the classification of institutions as public or private.

13 Bibliography

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